

The Strong Milman Question in arXiv:1912.10679 Was Answered Negatively in arXiv:2110.03023

Literature-status note

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Exact source question

Question 5.2 on PDF page 20 of Gowers–Wyczesany, *High-dimensional tennis balls* (arXiv:1912.10679), asks the following. Fix $k \in \mathbb{N}$, $C \geq 1$, and $\varepsilon > 0$. Does some dimension n guarantee that every norm $\|\cdot\|$ on $\mathbb{R}^n$ satisfying

$$\|x\|_2 \leq \|x\| \leq C\|x\|_2$$

has a k -dimensional subspace U for which, for some $\beta > 0$,

$$\beta\|u\|_2 \leq \|u\| \leq \beta(1 + \varepsilon)\|u\|_2 \quad (u \in U),$$

and the Euclidean orthogonal projection P_U has operator norm at most $1 + \varepsilon$ in $\|\cdot\|$? In the terminology of the follow-up, U must be both strongly $(1 + \varepsilon)$ -Euclidean and strongly $(1 + \varepsilon)$ -complemented. Source PDF page 21 announces that a follow-up will answer this question negatively.

The later answer

The announced follow-up is Gowers–Wyczesany, *A counterexample to a strengthening of a question of Milman* (arXiv:2110.03023). It restates the question as Question 1.3 and proves on PDF page 2:

Theorem 1.4. There are constants $\varepsilon > 0$ and C such that, for every sufficiently large n , an n -dimensional C -Euclidean normed space exists with no two-dimensional subspace that is both strongly $(1 + \varepsilon)$ -Euclidean and strongly $(1 + \varepsilon)$ -complemented.

Taking $k = 2$ directly negates the universal assertion in Question 5.2. The later paper even supplies explicit final constants (its concluding proof uses $\varepsilon = 2^{-1017}$), but the existence statement of Theorem 1.4 is already an exact answer.

Scope warning

Question 5.1 of arXiv:1912.10679 asks the weaker, coordinate-free Milman question using arbitrary near-isometries and arbitrary projections. The follow-up explicitly says it does not answer that weaker question. Therefore this identification closes Question 5.2 only; it makes no claim about Question 5.1 or the paper’s broader topological-subspace questions.

References

- [1] W. T. Gowers and K. Wyczesany, *High-dimensional tennis balls*, arXiv:1912.10679, Question 5.2.
- [2] W. T. Gowers and K. B. Wyczesany, *A counterexample to a strengthening of a question of Milman*, arXiv:2110.03023, Question 1.3 and Theorem 1.4.